CS 47N Personal Project

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Dataset:

I was interested in analyzing the Track & Field medalist results over the history of the Olympics (1896-2024). I collected the data from three different Kaggle files: a large dataset from years 1896-2016 and smaller datasets for Tokyo 2020 and Paris 2024. The dataset includes columns for the year, location, country, athlete, gender, event, result, and medal.

e.g.

year	location	country	name	gender	event	result	sec	medal
2020	Tokyo	ETH	SELEMON BAREGA	M	Men's 10,000m	27:43.22	1663.22	G
2020	Tokyo	UGA	JOSHUA CHEPTEGEI	M	Men's 10,000m	27:43.63	1663.63	S
2020	Tokyo	UGA	JACOB KIPLIMO	M	Men's 10,000m	27:43.88	1663.88	B

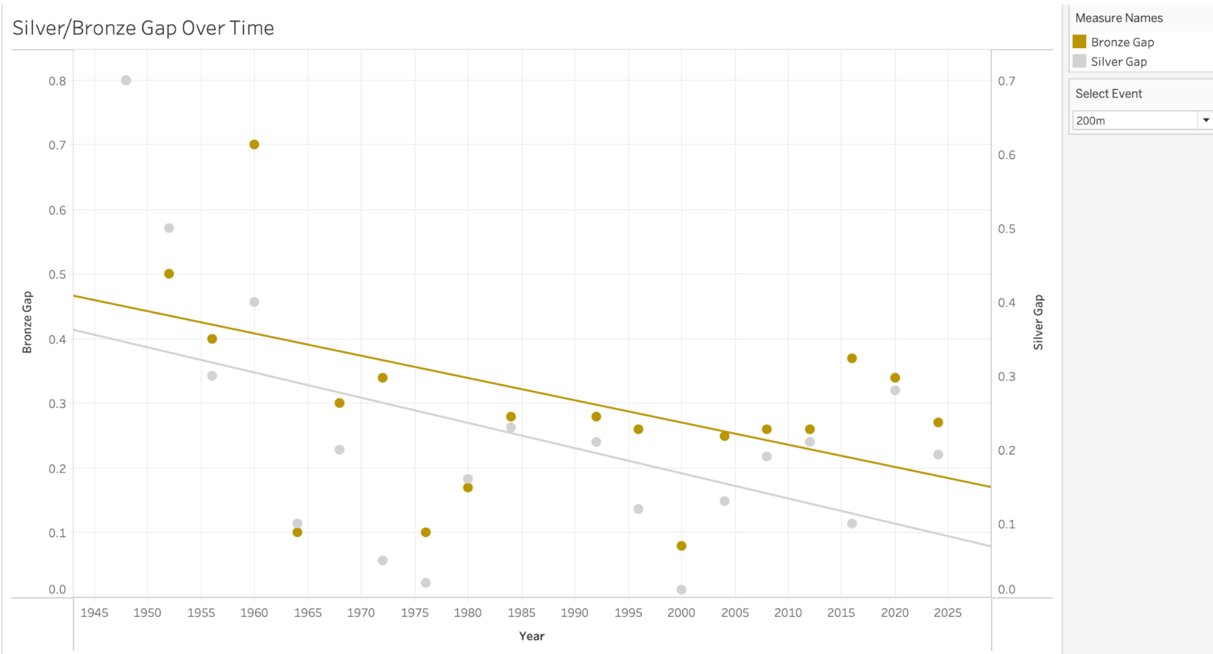
Data preparation:

I combined the three datasets using Tableau Prep. For the Tokyo and Paris datasets, I had to combine the results file with the medals file using the join feature. Before using a union between the datasets, I removed irrelevant columns and standardized naming conventions for athlete name, medal, and event. I had to create a special new column "sec" to standardize the "result" column, as I realized during later analysis that numerical values were processed differently based on whether a colon or period was used.

Questions and answers:

1. Is Track & Field becoming more competitive? i.e. gap between gold/silver/bronze?

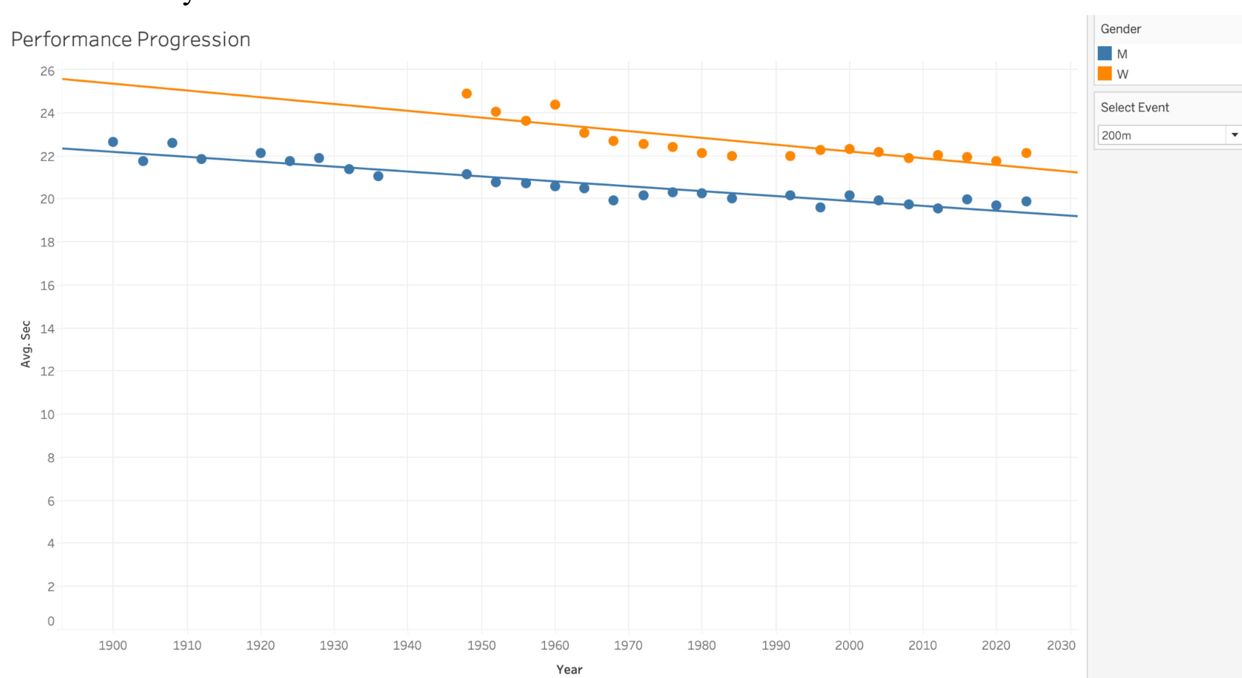
Tableau: created two new fields "Silver Gap" and "Gold Gap" that calculated the difference between the winning gold result and the silver/bronze results for each year. Scatterplot visualization with regression lines filtered by event.



Conclusion: As a whole, Track & Field performance in the Olympics is becoming more competitive—but there are varying results by event.

2. How has Track & Field performance progressed since the beginning of the Olympics? How has the performance gap between men and women changed over time?

Tableau: Plotted year vs. sec, colored by gender. Scatterplot visualization with regression lines filtered by event.



Conclusion: Overall performance in Track & Field in the Olympics has improved, signaling the constant improvement in the sport. For most events, the gap between men and women seems to remain relatively consistent over time.

3. Which countries specialize in which events?

[Pandas](#): First looked at overall medal count grouping by country, then filtered to distance/sprint/field events only, then added filters for the last 50 years to look at any recent trends.

```python
# top 5 countries in distance events
non_distance_events = ['High Jump', 'Long Jump', 'Triple Jump', 'Pole Vault',
                'Shot Put', 'Discus', 'Hammer', 'Javelin',
                'Decathlon', 'Heptathlon', '100m', '200m','400m','800m']

# Filter to distance events only
distance = table[~table.event.str.contains('|'.join(non_distance_events), case=False)]
distance.groupby('country').medal.count().sort_values(ascending=False).head(5)
```

country	medal
USA	118
KEN	86
ETH	57
GBR	53
FIN	47

```python
# top 5 countries in sprint events
non_sprint_events = ['High Jump', 'Long Jump', 'Triple Jump', 'Pole Vault',
                'Shot Put', 'Discus', 'Hammer', 'Javelin',
                'Decathlon', 'Heptathlon', '1500m', '3000m','10,000m','Marathon','20km','50km']

# Filter to sprint events only
sprint = table[~table.event.str.contains('|'.join(non_sprint_events), case=False)]
sprint.groupby('country').medal.count().sort_values(ascending=False).head(5)
```

country	medal
USA	433
GBR	109
JAM	96
KEN	67
URS	42

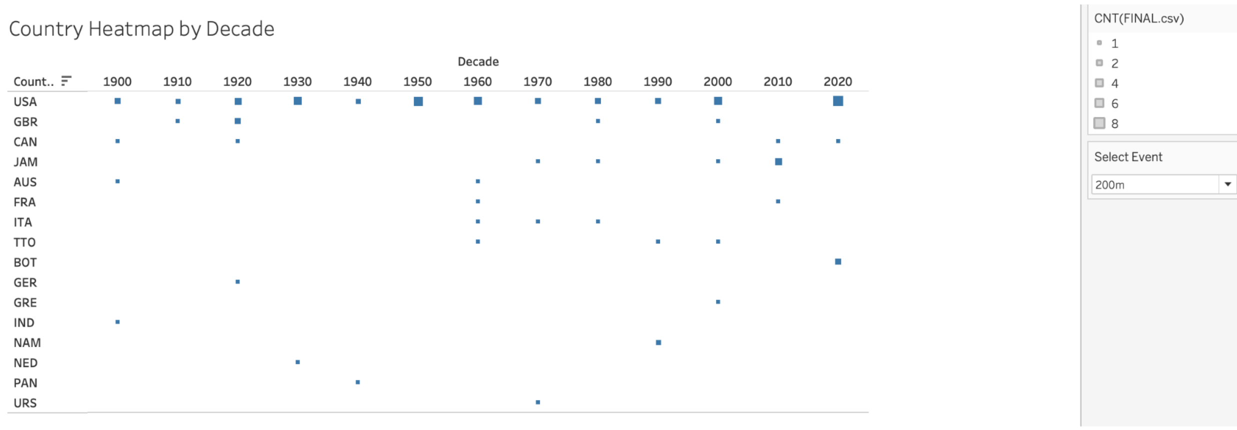
```python
# top 5 countries in field events
field_events = ['High Jump', 'Long Jump', 'Triple Jump', 'Pole Vault',
                'Shot Put', 'Discus', 'Hammer', 'Javelin',
                'Decathlon', 'Heptathlon']

# Filter to field events only
field = table[table.event.str.contains('|'.join(field_events), case=False)]
field.groupby('country').medal.count().sort_values(ascending=False).head(5)
```

country	medal
USA	254
URS	68
GER	61
RUS	37
GBR	34

Conclusion: While the U.S. is overall the most dominant country in Track & Field medalists, it has lagged behind Kenya and Ethiopia in distance events (1500m, 5000m, 10,000m, Marathon, etc).

4. Adding onto Question 3, how has country specialization changed over time?

Tableau: Bucketed years by decade (chunks of 10) then plotted medal count for each country in heatmap, sorted by descending overall medal count.



Conclusion: Found similar trends in the last 50 years as those in quantitative analysis from Question 3, as more countries become competitive in Olympic Track & Field. Interestingly, the U.S. Men's 4 x 100m team has seen no medals in the last 20 years, despite having some of the top performers in the 100m during that time period.

5. Which event has seen the most relative improvement since the first Olympics it appeared in?

Pandas: Filtered to only look at gold medalists, then compared relative improvement from the first Olympics the event appeared in to most recent (especially relevant since results from women did not appear until the mid 20th-century). Sorted table in descending order by relative improvement.

event	first_year	first_result	last_year	last_result	pct_improvement
Men's Marathon	1904.0	12533.00	2024.0	7586.00	39.47
Men's 110m Hurdles	1896.0	17.60	2024.0	12.99	26.19
Men's 50km Race Walk	1932.0	17410.00	2016.0	13258.00	23.85
Men's 800m	1896.0	131.00	2024.0	101.19	22.76
Men's 1500m	1896.0	273.20	2024.0	212.54	22.20
Men's 400m Hurdles	1900.0	57.60	2024.0	46.46	19.34
Men's 400m	1896.0	54.20	2024.0	44.28	18.30
Men's 100m	1896.0	12.00	2024.0	9.83	18.08
Men's 4 x 400m Relay	1908.0	209.40	2020.0	175.70	16.09
Women's 4 x 100m Relay	1928.0	48.40	2020.0	41.02	15.25
Men's 10,000m	1912.0	1880.80	2024.0	1603.14	14.76
Women's 800m	1928.0	136.80	2024.0	116.86	14.58
Men's 20km Race Walk	1956.0	5487.40	2016.0	4754.00	13.37
Men's 4 x 100m Relay	1912.0	42.40	2020.0	37.50	11.56
Women's 100m	1928.0	12.20	2024.0	10.84	11.15
Women's 200m	1948.0	24.40	2024.0	21.83	10.53
Men's 5000m	1912.0	876.60	2024.0	793.66	9.46
Men's 200m	1900.0	22.20	2024.0	20.10	9.46
Women's 400m Hurdles	1984.0	54.61	2024.0	50.37	7.76
Women's 400m	1964.0	52.00	2024.0	48.17	7.37
Women's 1500m	1972.0	241.38	2024.0	231.29	4.18
Women's 4 x 400m Relay	1972.0	203.00	2020.0	196.85	3.03
Women's Marathon	1984.0	8692.00	2024.0	8575.00	1.35
Women's 10,000m	1992.0	1866.02	2024.0	1843.25	1.22
Women's 100m Hurdles	1972.0	12.59	2024.0	12.53	0.48
Men's 3000m Steeplechase	2020.0	488.90	2020.0	488.90	0.00
Women's 5000m	1996.0	899.88	2024.0	900.73	-0.09
Women's 3000m Steeplechase	2008.0	538.81	2024.0	555.11	-3.03
Women's 20km Race Walk	2012.0	5102.00	2016.0	5315.00	-4.17
Men's 3000 Steeplechase	1900.0	454.40	2016.0	483.28	-6.36

Conclusion: The Men's Marathon has shown the greatest relative improvement from its first appearance in the Olympics to the most recent result. Distance events tend to show greater improvement, likely since they are longer timed events. Interestingly, some results show a negative percent. This is due to the limitations of my calculations that only base relative improvement on the two extreme ends of results, susceptible to outlier performances and ignoring performances between the first and last years.

<u>Lessons learned:</u>

The most difficult part of this project was data cleaning and preparation. In combining the three datasets, I faced many challenges in formatting inconsistencies and missing/irrelevant columns. To my surprise, this ended up being the most time-consuming part of the project. I kept having to return to my Tableau Prep workspace to sort out additional kinks I found during the analysis process. Once I had the cleaned dataset, the analysis and visualization were easier (and more interesting!). Specifically for this sport, the analysis for running versus field events was inherently distinct—greater values for field sports is better versus lower values for running. Thus, to answer many of my analysis and visualization questions, I had to make note of which kind of event I was looking at and adjust my calculations accordingly. To conclude, my biggest takeaway was the value of corroborating my findings using different analysis tools. As someone who has watched the past few Track & Field Olympics closely, I had a feel for some of the data. In observing with a critical eye, I saw how a single visualization or number fails to capture the whole story. Using techniques in both Tableau and Pandas captured more nuance and allowed me to feel much more confident in my overall findings.

<u>Next steps:</u>

As a prospective Data Science major and runner myself, I plan to continue on my "datathletics" journey. Academically, I would love to continue to learn more skills specific for sports analytics (and would appreciate any class recommendations!). Hearing from the sports coaches and data specialists for the Giants was super eye-opening to a potential career that I would like to explore. On the personal side, I'm curious about adopting some of my data analysis/visualizations from this project for my own running data. My dad and I are training for a marathon for the end of the summer, and I'm interested in uploading all of the data I have from my Garmin to see any interesting trends. We're both datanerds and are always looking for any edge!